Num Anal Homework-4

1. A small community has a population of 400, of which, at time 0, 201 are infected with a disease and thirteen are vaccinated against the disease. The infected are well mixed within the community. Write down an appropriate disease model. If the rate of infection is 0.001 and the rate of recovery is 0.1, show that the disease will grow. When the disease epidemic is at its highest point, how many people are infected? At that time, how many susceptible persons are there? Finally, when the disease dies out, are there any susceptible persons left? Give your answer and show how you arrived at the answer.

2. Find the solution P_n using the given initial value P_0 of the following models.

   Model 1: $P_{n+1} = 7 P_n / (3 P_n + 1)$, initial value $P_0 = 1$.

   Model 2: $P_{n+1} = (P_n - 5)^2 + 5$, initial value $P_0 = 6$.